

Cyber Forensics - 24CY611

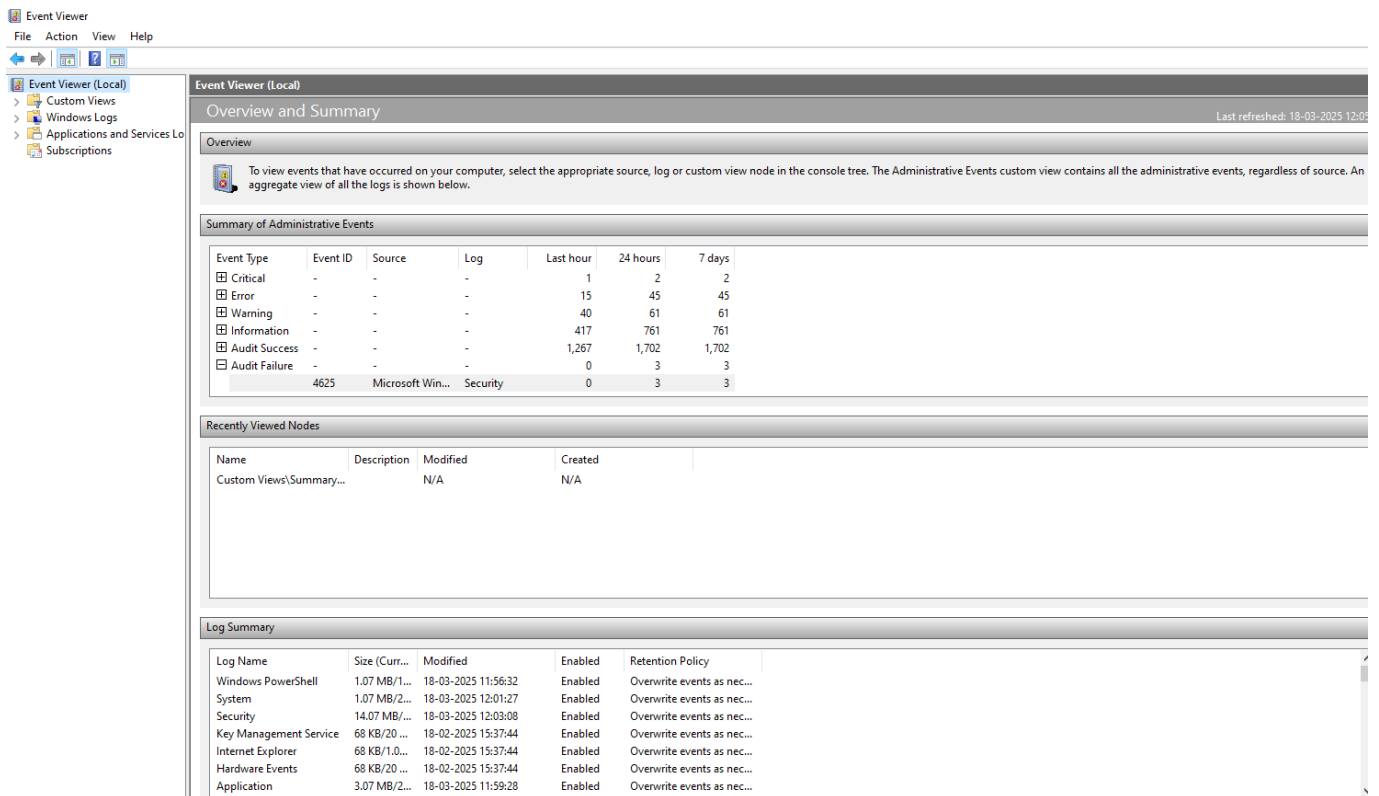
Lab 7 - Network Forensics

1. Capturing and Analyzing the logs of a Computer using windows event viewer

Open the Event Viewer application on your Windows computer.

In the left pane, locate and expand "Event Viewer (Local)" if it is not already expanded.

Expand "Windows Logs" to view the available log categories.

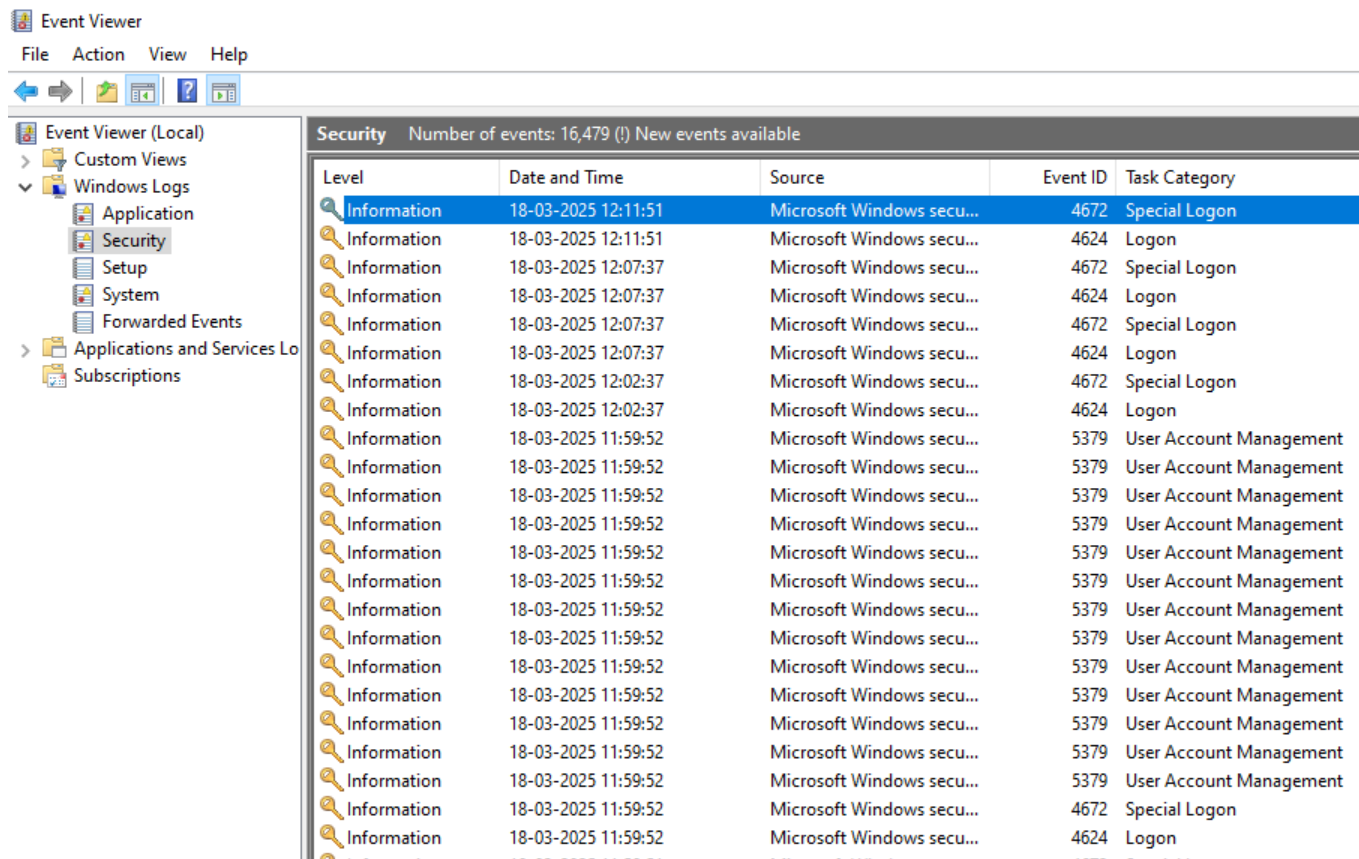


The screenshot shows the Windows Event Viewer application. The left pane displays the tree structure with "Event Viewer (Local)" expanded, showing "Custom Views", "Windows Logs", "Applications and Services Logs", and "Subscriptions". The main pane shows the "Overview and Summary" page for the "Security" log. The "Summary of Administrative Events" table is visible, showing event counts for various types. The "Recently Viewed Nodes" table shows a single entry for "Custom Views/Summary...". The "Log Summary" table at the bottom lists various logs and their properties.

Event Type	Event ID	Source	Log	Last hour	24 hours	7 days
Critical	-	-	-	1	2	2
Error	-	-	-	15	45	45
Warning	-	-	-	40	61	61
Information	-	-	-	417	761	761
Audit Success	-	-	-	1,267	1,702	1,702
Audit Failure	-	-	-	0	3	3
	4625	Microsoft Win...	Security	0	3	3

Name	Description	Modified	Created
Custom Views/Summary...	N/A	N/A	N/A

Log Name	Size (Curr...	Modified	Enabled	Retention Policy
Windows PowerShell	1.07 MB/1...	18-03-2025 11:56:32	Enabled	Overwrite events as nec...
System	1.07 MB/2...	18-03-2025 12:01:27	Enabled	Overwrite events as nec...
Security	14.07 MB/...	18-03-2025 12:03:08	Enabled	Overwrite events as nec...
Key Management Service	68 KB/20 ...	18-02-2025 15:37:44	Enabled	Overwrite events as nec...
Internet Explorer	68 KB/1.0...	18-02-2025 15:37:44	Enabled	Overwrite events as nec...
Hardware Events	68 KB/20 ...	18-02-2025 15:37:44	Enabled	Overwrite events as nec...
Application	3.07 MB/2...	18-03-2025 11:59:28	Enabled	Overwrite events as nec...



Click on a specific event in the list. The event's summary will appear in the lower pane.

The screenshot shows the Windows Event Viewer interface. On the left, the 'Security' log is selected under 'Windows Logs'. The main pane displays a list of events. Event 4624 is selected, and its details are shown in the lower pane.

Level	Date and Time	Source	Event ID	Task Category
Information	18-03-2025 12:11:51	Microsoft Windows secu...	4672	Special Logon
Information	18-03-2025 12:11:51	Microsoft Windows secu...	4624	Logon
Information	18-03-2025 12:07:37	Microsoft Windows secu...	4672	Special Logon
Information	18-03-2025 12:07:37	Microsoft Windows secu...	4624	Logon
Information	18-03-2025 12:07:37	Microsoft Windows secu...	4672	Special Logon
Information	18-03-2025 12:07:37	Microsoft Windows secu...	4624	Logon

Event 4624, Microsoft Windows security auditing.

General Details

An account was successfully logged on.

Subject:

- Security ID: SYSTEM
- Account Name: DESKTOP-8JISE8QS
- Account Domain: WORKGROUP
- Logon ID: 0x3E7

Logon Information:

- Logon Type: 5
- Restricted Admin Mode: -
- Virtual Account: No
- Elevated Token: Yes

Impersonation Level: Impersonation

New Logon:

- Security ID: SYSTEM
- Account Name: SYSTEM
- Account Domain: NT AUTHORITY
- Logon ID: 0x3E7
- Linked Logon ID: 0x0
- Network Account Name: -
- Network Account Domain: -
- Logon GUID: {00000000-0000-0000-0000-000000000000}

Log Name: Security

Source: Microsoft Windows security

Event ID: 4624

Level: Information

User: N/A

OpCode: Info

More Information: [Event Log Online Help](#)

To see the full details of the selected event, click "**Details**" or "**General**" in the event properties pane. This section will display detailed information such as the event ID, source, and description.

The screenshot shows the Windows Event Viewer interface. On the left, the 'Application' log is selected under 'Windows Logs'. The main pane displays a list of events. Event 16384 is selected, and its details are shown in the lower pane.

Level	Date and Time	Source	Event ID	Task Category
Information	18-03-2025 12:13:01	Security-SPP	16384	None
Information	18-03-2025 12:12:24	Security-SPP	16394	None
Information	18-03-2025 12:08:32	Security-SPP	16384	None
Information	18-03-2025 12:08:12	SecurityCenter	15	None
Information	18-03-2025 12:07:58	SecurityCenter	15	None
Information	18-03-2025 12:07:33	Security-SPP	16394	None
Information	18-03-2025 12:07:30	SecurityCenter	15	None
Information	18-03-2025 11:59:17	Security-SPP	16384	None
Information	18-03-2025 11:59:01	SecurityCenter	15	None
Information	18-03-2025 11:58:59	SecurityCenter	1	None
Information	18-03-2025 11:58:58	edgeupdate	0	None
Information	18-03-2025 11:58:50	Security-SPP	16394	None
Information	18-03-2025 11:58:24	Security-SPP	903	None
Information	18-03-2025 11:58:24	Security-SPP	16384	None

Event 16384, Security-SPP

General Details

Successfully scheduled Software Protection service for re-start at 2025-03-19T06:27:32Z. Reason: RulesEngine.

Log Name: Application

Source: Security-SPP

Event ID: 16384

Level: Information

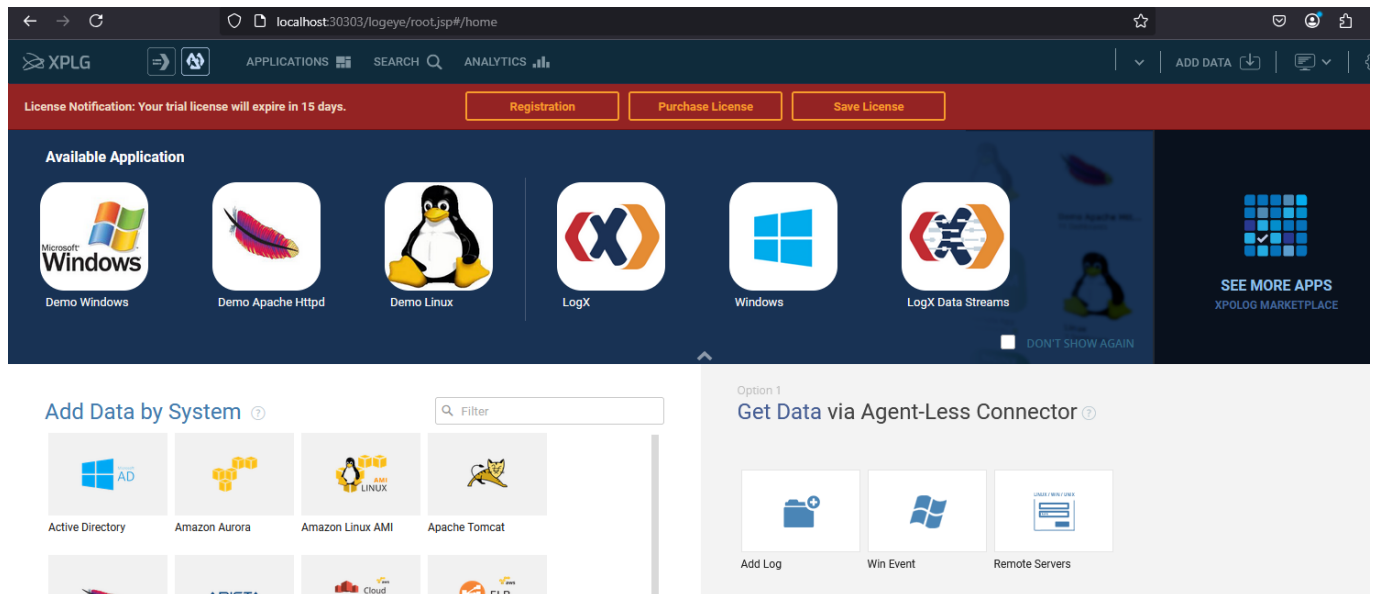
User: N/A

OpCode: Info

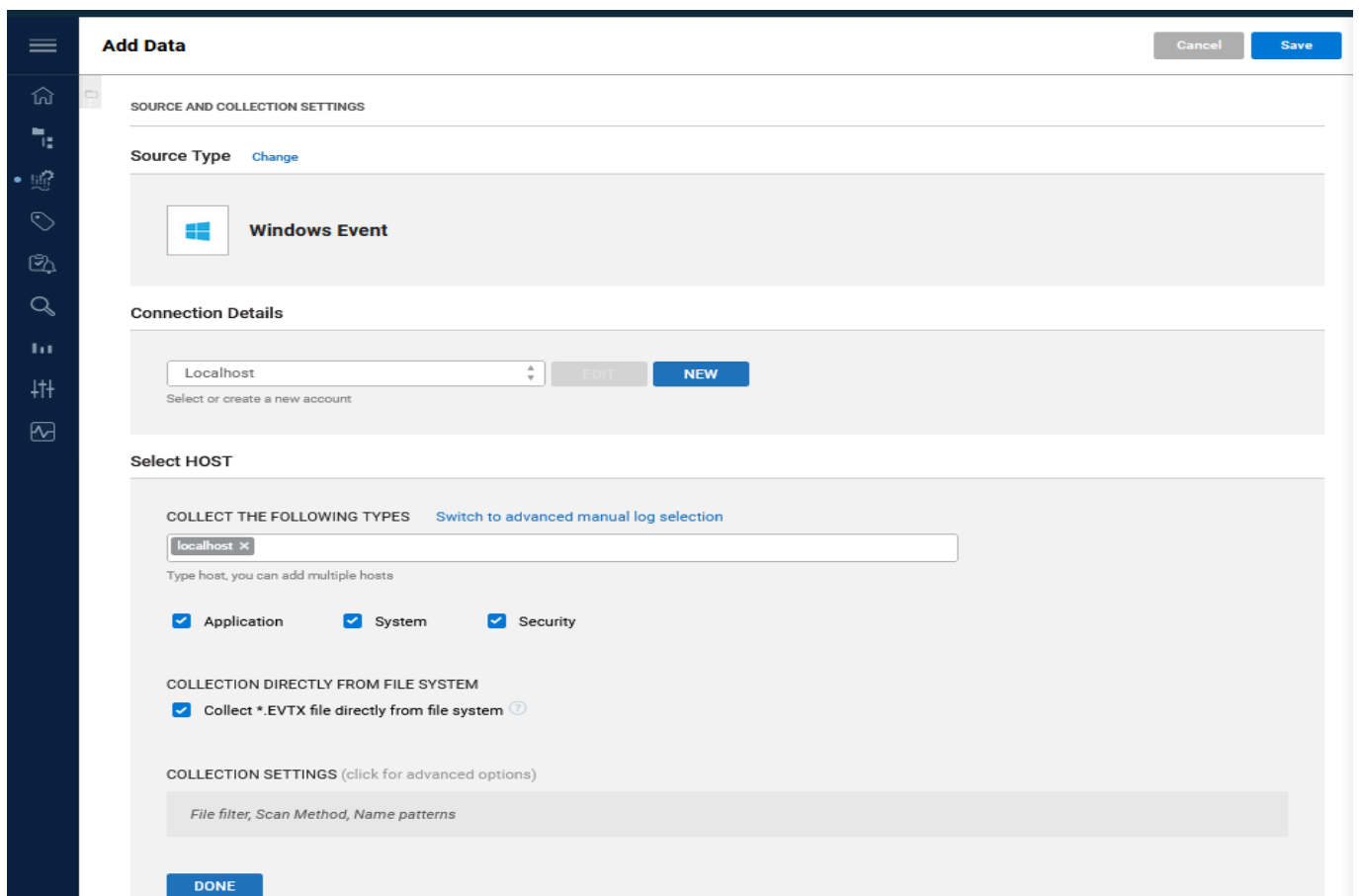
More Information: [Event Log Online Help](#)

2. Investigating System Log Data Using XpoLog Center Suite Tool

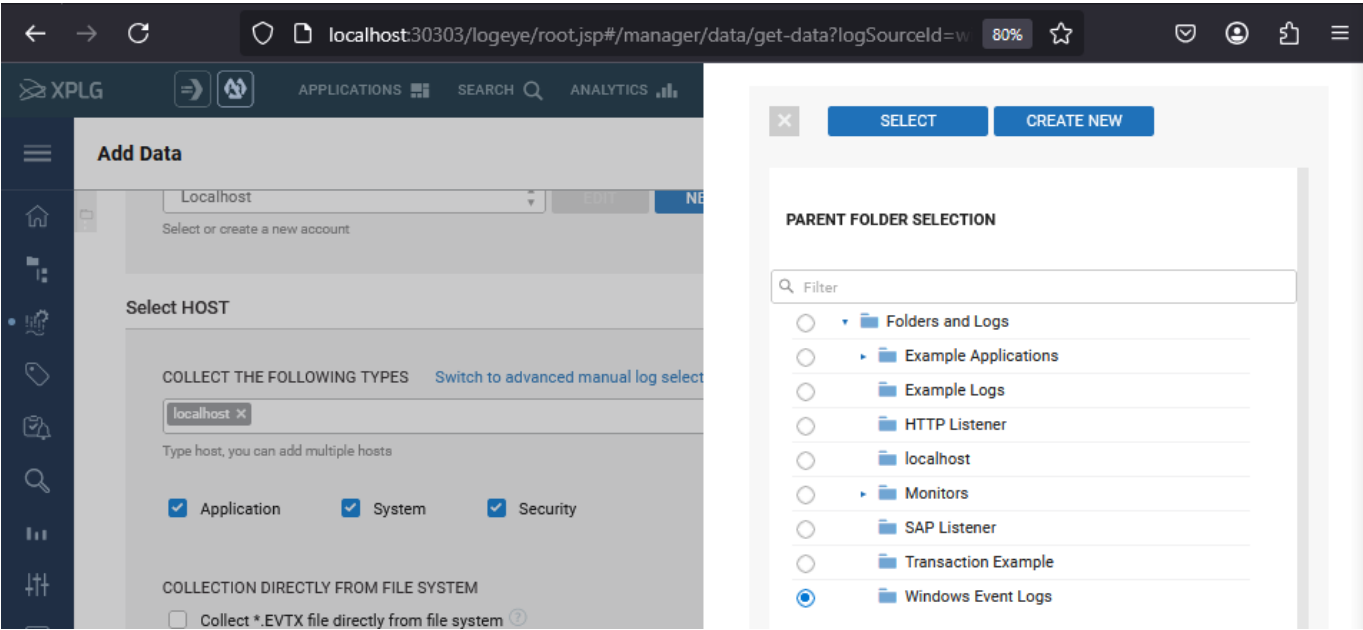
Open XpoLog GUI in the default web browser (URL: <http://localhost:30303/logeye/root.html>), click Win Event



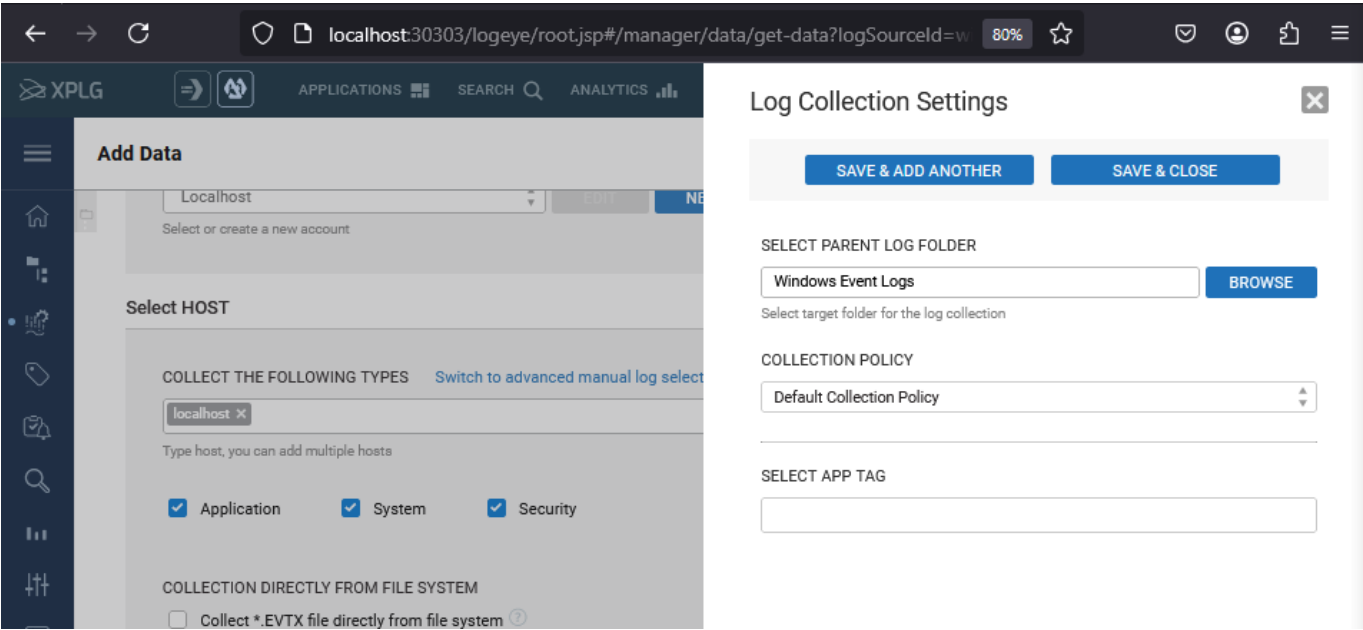
Click save



Select the Windows Event Logs from Parent folder selection



Then click save and close



Folders and Logs						
Add ▾		Actions ▾		Filter		
☐	Name	Description	AppTag	Collection Policy	Last Collected	Size
☐	▶ Example Applications	...				
☐	▶ Example Logs	...	Example AppTag			
☐	Example HTTP Listener	...				
☐	▶ localhost	...				
☐	▶ Monitors	...				
☐	Example SAP Listener	...				
☐	▶ Transaction Example	...	Example AppTag			
☒	▼ Windows Event Logs	...	Windows Event Logs			
☒	Application	Windows application events	Windows Event Logs	Default Collection Policy	03/18/2025 14:29:00	1.2 MB
☒	Security	Windows security events	Windows Event Logs	Default Collection Policy	03/18/2025 14:29:00	12.9 MB
☒	System	Windows system events	Windows Event Logs	Default Collection Policy	03/18/2025 14:16:00	465.6 KB

Log Viewer

Application

Security

25

25 / 25 records

FILTER

All Time

Example Applications

Example Logs

HTTP Listener

localhost

Monitors

SAP Listener

Transaction Example

Windows Event Logs

Application

Security

System

Filtering

Type

Date

Account Name

Account Domain

Source

Category

Event

User

Con

Information

03/18/2025 14:11:20

Software Protection Platform Service

None

1003

DESI

Error

03/18/2025 14:11:20

Software Protection Platform Service

None

8198

DESI

Information

03/18/2025 14:11:53

Software Protection Platform Service

None

16384

DESI

Success

03/18/2025 14:11:53

Software Protection Platform Service

None

903

DESI

Information

03/18/2025 14:13:21

edgeupdate

None

0

DESI

Information

03/18/2025 14:13:22

Software Protection Platform Service

None

900

DESI

Information

03/18/2025 14:13:22

Software Protection Platform Service

None

16394

DESI

Information

03/18/2025 14:13:23

SecurityCenter

None

1

DESI

Information

03/18/2025 14:13:25

SecurityCenter

None

15

DESI

Information

03/18/2025 14:13:36

Wicintfy

None

6003

DESI

Information

03/18/2025 14:13:37

Wicintfy

None

6000

DESI

Information

03/18/2025 14:13:38

Software Protection Platform Service

None

1003

DESI

Error

03/18/2025 14:13:38

Software Protection Platform Service

None

8198

DESI

Information

03/18/2025 14:13:57

SecurityCenter

None

15

DESI

Information

03/18/2025 14:14:35

Software Protection Platform Service

None

16384

DESI

Information

03/18/2025 14:17:17

Software Protection Platform Service

None

16394

DESI

Information

03/18/2025 14:18:12

Software Protection Platform Service

None

16384

DESI

Information

03/18/2025 14:21:29

Software Protection Platform Service

None

16394

DESI

Information

03/18/2025 14:21:30

SecurityCenter

None

15

DESI

Information

03/18/2025 14:21:59

Software Protection Platform Service

None

16384

DESI

Information

03/18/2025 14:23:57

SecurityCenter

None

15

DESI

Information

03/18/2025 14:27:35

Software Protection Platform Service

None

16394

DESI

Information

03/18/2025 14:28:07

Software Protection Platform Service

None

16384

DESI

Information

03/18/2025 14:28:40

Software Protection Platform Service

None

16394

DESI

Information

03/18/2025 14:29:10

Software Protection Platform Service

None

16384

DESI

To view Windows Security logs, click Security under Windows Event Logs. All the Security related logs will appear as shown in the following screenshot:

Log Viewer

Folders and Logs

Example Applications

Example Logs

HTTP Listener

localhost

Monitors

SAP Listener

Transaction Example

Windows Event Logs

Application

Security

System

Security		Application										
25												
FILTERING												
Type	Date	Account Name	Account Domain	Source	Category	Event	Event Description	Category Description	Sub Category	User	Logon ID	Cc
Audit Success	03/18/2025 14:13:55	SYSTEM	NT	Microsoft-Windows-Security-Auditing	12548	4672	Special privileges assigned to new logon	Privilege Use	Sensitive Privilege Use / Non-Sensitive Privilege Use		0x3e7	DE
Audit Success	03/18/2025 14:13:56	DESKTOP-BJSEBQ	WORKGROUP	Microsoft-Windows-Security-Auditing	12544	4634	An account was successfully logged on	Logon/Logoff	Logon		0x3e7	DE
Audit Success	03/18/2025 14:13:56	SYSTEM	NT	Microsoft-Windows-Security-Auditing	12548	4672	Special privileges assigned to new logon	Privilege Use	Sensitive Privilege Use / Non-Sensitive Privilege Use		0x3e7	DE
Audit Success	03/18/2025 14:13:56	DESKTOP-BJSEBQ	WORKGROUP	Microsoft-Windows-Security-Auditing	13824	5382	Vault credentials were read				0x3e7	DE
Audit Success	03/18/2025 14:17:09	DESKTOP-BJSEBQ	WORKGROUP	Microsoft-Windows-Security-Auditing	12544	4624	An account was successfully logged on	Logon/Logoff	Logon		0x3e7	DE
Audit Success	03/18/2025 14:17:09	SYSTEM	NT	Microsoft-Windows-Security-Auditing	12548	4672	Special privileges assigned to new logon	Privilege Use	Sensitive Privilege Use / Non-Sensitive Privilege Use		0x3e7	DE
Audit Success	03/18/2025 14:17:17	DESKTOP-BJSEBQ	WORKGROUP	Microsoft-Windows-Security-Auditing	12544	4634	An account was successfully logged on	Logon/Logoff	Logon		0x3e7	DE
Audit Success	03/18/2025 14:17:17	SYSTEM	NT	Microsoft-Windows-Security-Auditing	12548	4672	Special privileges assigned to new logon	Privilege Use	Sensitive Privilege Use / Non-Sensitive Privilege Use		0x3e7	DE
Audit Success	03/18/2025 14:18:37	DESKTOP-BJSEBQ	WORKGROUP	Microsoft-Windows-Security-Auditing	12544	4634	An account was successfully logged on	Logon/Logoff	Logon		0x3e7	DE
Audit Success	03/18/2025 14:18:37	SYSTEM	NT	Microsoft-Windows-Security-Auditing	12548	4672	Special privileges assigned to new logon	Privilege Use	Sensitive Privilege Use / Non-Sensitive Privilege Use		0x3e7	DE
Audit Success	03/18/2025 14:21:28	DESKTOP-BJSEBQ	WORKGROUP	Microsoft-Windows-Security-Auditing	12544	4634	An account was successfully logged on	Logon/Logoff	Logon		0x3e7	DE
Audit Success	03/18/2025 14:21:28	SYSTEM	NT	Microsoft-Windows-Security-Auditing	12548	4672	Special privileges assigned to new logon	Privilege Use	Sensitive Privilege Use / Non-Sensitive Privilege Use		0x3e7	DE
Audit Success	03/18/2025 14:23:55	DESKTOP-BJSEBQ	WORKGROUP	Microsoft-Windows-Security-Auditing	12544	4634	An account was successfully logged on	Logon/Logoff	Logon		0x3e7	DE
Audit Success	03/18/2025 14:23:55	SYSTEM	NT	Microsoft-Windows-Security-Auditing	12548	4672	Special privileges assigned to new logon	Privilege Use	Sensitive Privilege Use / Non-Sensitive Privilege Use		0x3e7	DE
Audit Success	03/18/2025 14:23:55	DESKTOP-BJSEBQ	WORKGROUP	Microsoft-Windows-Security-Auditing	13824	5382	Vault credentials were read				0x3e7	DE
Audit Success	03/18/2025 14:26:15	DESKTOP-BJSEBQ	WORKGROUP	Microsoft-Windows-Security-Auditing	12544	4634	An account was successfully logged on	Logon/Logoff	Logon		0x3e7	DE
Audit Success	03/18/2025 14:26:15	SYSTEM	NT	Microsoft-Windows-Security-Auditing	12548	4672	Special privileges assigned to new logon	Privilege Use	Sensitive Privilege Use / Non-Sensitive Privilege Use		0x3e7	DE
Audit Success	03/18/2025 14:27:35	DESKTOP-BJSEBQ	WORKGROUP	Microsoft-Windows-Security-Auditing	12544	4634	An account was successfully logged on	Logon/Logoff	Logon		0x3e7	DE
Audit Success	03/18/2025 14:27:35	SYSTEM	NT	Microsoft-Windows-Security-Auditing	12548	4672	Special privileges assigned to new logon	Privilege Use	Sensitive Privilege Use / Non-Sensitive Privilege Use		0x3e7	DE
Audit Success	03/18/2025 14:27:37	DESKTOP-BJSEBQ	WORKGROUP	Microsoft-Windows-Security-Auditing	12544	4634	An account was successfully logged on	Logon/Logoff	Logon		0x3e7	DE
Audit Success	03/18/2025 14:27:37	SYSTEM	NT	Microsoft-Windows-Security-Auditing	12548	4672	Special privileges assigned to new logon	Privilege Use	Sensitive Privilege Use / Non-Sensitive Privilege Use		0x3e7	DE
Audit Success	03/18/2025 14:28:37	DESKTOP-BJSEBQ	WORKGROUP	Microsoft-Windows-Security-Auditing	12544	4634	An account was successfully logged on	Logon/Logoff	Logon		0x3e7	DE
Audit Success	03/18/2025 14:28:37	SYSTEM	NT	Microsoft-Windows-Security-Auditing	12548	4672	Special privileges assigned to new logon	Privilege Use	Sensitive Privilege Use / Non-Sensitive Privilege Use		0x3e7	DE
Audit Success	03/18/2025 14:28:38	DESKTOP-BJSEBQ	WORKGROUP	Microsoft-Windows-Security-Auditing	12544	4634	An account was successfully logged on	Logon/Logoff	Logon		0x3e7	DE
Audit Success	03/18/2025 14:28:38	SYSTEM	NT	Microsoft-Windows-Security-Auditing	12548	4672	Special privileges assigned to new logon	Privilege Use	Sensitive Privilege Use / Non-Sensitive Privilege Use		0x3e7	DE

To view Windows System logs, click System under Windows Event Logs. All the System-related logs will appear as shown in the following screenshot

Log Viewer

Folders and Logs

Example Applications

Example Logs

HTTP Listener

localhost

Monitors

SAP Listener

Transaction Example

Windows Event Logs

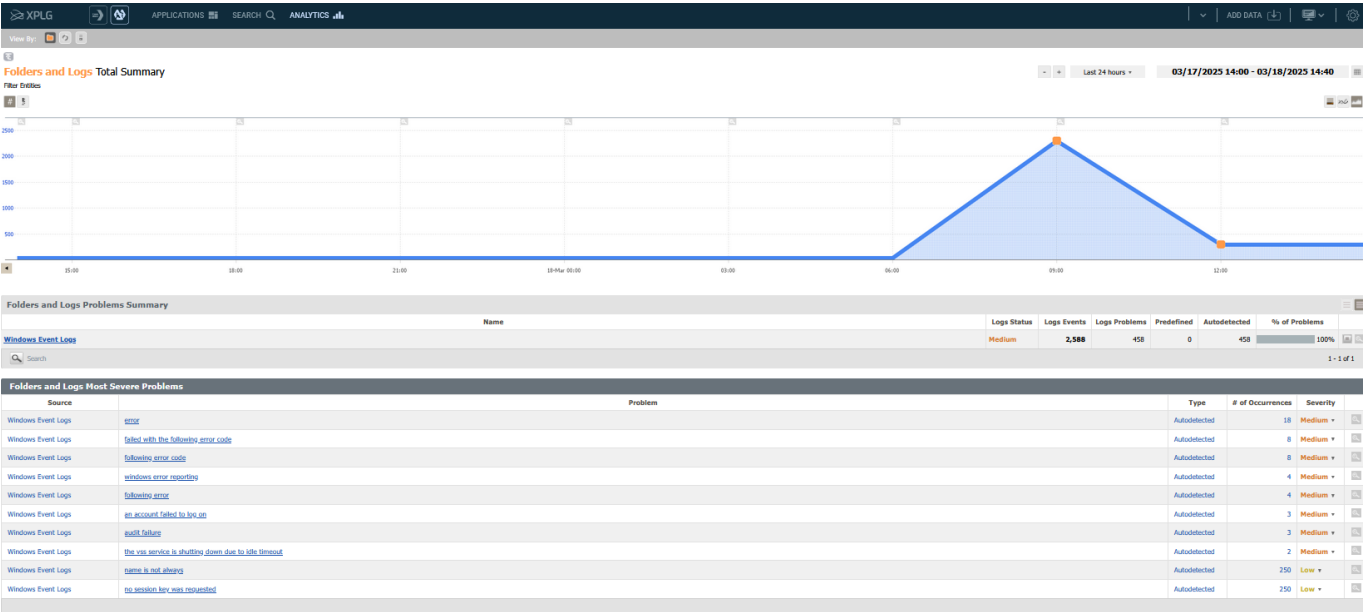
Application

Security

System

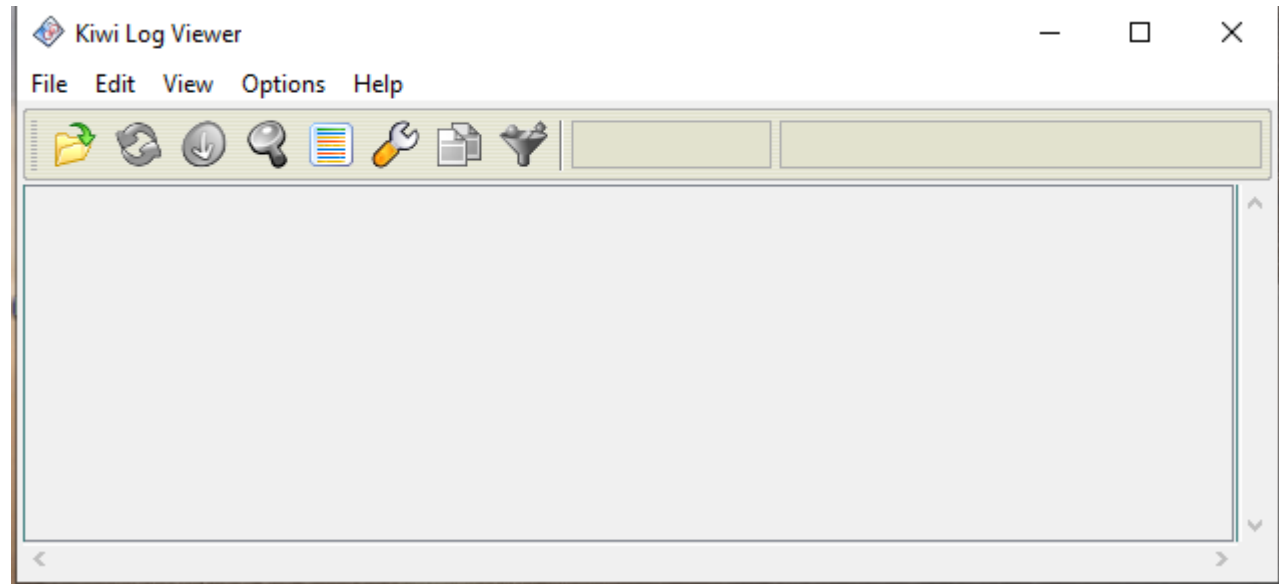
System		Security		Application					
25	25 / 25 records								
FILTERING									
Type	Date	Account Name	Account Domain	Source	Category	Event	User	Computer	Description
Information	03/18/2025 14:11:12			Microsoft-Windows-Directory-Services-SAM	None	16962	SYSTEM	DESKTOP-BJSEBQ	Remote calls to the SAM database are being restricted using the default security descriptor: O:SYG:...
Information	03/18/2025 14:11:12			Microsoft-Windows-Directory-Services-SAM	None	16977	SYSTEM	DESKTOP-BJSEBQ	For more information please see http://go.microsoft.com/fwlink/?LinkID=787651 .
Information	03/18/2025 14:11:12			Microsoft-Windows-Directory-Services-SAM	None	16983	SYSTEM	DESKTOP-BJSEBQ	The domain is configured with the following minimum password length-related settings.
Information	03/18/2025 14:11:13			Microsoft-Windows-Dhcp-Client	4	50036	LOCAL SERVICE	DESKTOP-BJSEBQ	The security account manager is now logging periodic summary events for remote clients that ca...
Information	03/18/2025 14:11:13			Microsoft-Windows-Dhcp-Client	4	50103	LOCAL SERVICE	DESKTOP-BJSEBQ	DHCPv4 client service is started
Information	03/18/2025 14:11:13			Microsoft-Windows-DHCPv6-Client	4	51046	LOCAL SERVICE	DESKTOP-BJSEBQ	DHCPv6 client service is started
Information	03/18/2025 14:11:13			Microsoft-Windows-FilterManager	None	6	SYSTEM	DESKTOP-BJSEBQ	File System Filter 'wcfis' (10.0, 2047-08-27T10:58:50.0000000Z) has successfully loaded and registere...
Information	03/18/2025 14:11:13			Microsoft-Windows-FilterManager	None	6	SYSTEM	DESKTOP-BJSEBQ	File System Filter 'luafv' (10.0, 2091-08-26T23:39:08.0000000Z) has successfully loaded and registere...
Information	03/18/2025 14:11:13			Microsoft-Windows-FilterManager	None	6	SYSTEM	DESKTOP-BJSEBQ	File System Filter 'CidFtr' (10.0, 2039-01-26T03:11:47.0000000Z) has successfully loaded and register...
Information	03/18/2025 14:11:13			Microsoft-Windows-FilterManager	None	1	SYSTEM	DESKTOP-BJSEBQ	File System Filter 'CidFtr' (version 10.0, 2039-01-26T03:11:47.0000000Z) unloaded successfully.
Information	03/18/2025 14:11:13			Microsoft-Windows-FilterManager	None	6	SYSTEM	DESKTOP-BJSEBQ	File System Filter 'CidFtr' (10.0, 2039-01-26T03:11:47.0000000Z) has successfully loaded and regist...
Information	03/18/2025 14:11:13			Microsoft-Windows-FilterManager	None	6	SYSTEM	DESKTOP-BJSEBQ	File System Filter 'storosoff' (10.0, 2010-08-23T18:33:45.0000000Z) has successfully loaded and regi...
Information	03/18/2025 14:11:13			Microsoft-Windows-FilterManager	None	6	SYSTEM	DESKTOP-BJSEBQ	File System Filter 'bindfr' (10.0, 2057-01-09T03:38:59.0000000Z) has successfully loaded and regist...
Information	03/18/2025 14:11:15			Service Control Manager	None	7026	SYSTEM	DESKTOP-BJSEBQ	The following boot-start or system-start driver(s) did not load.
Error	03/18/2025 14:13:22			Service Control Manager	None	7023	SYSTEM	DESKTOP-BJSEBQ	The System Guard Runtime Monitor Broker service terminated with the following error:
Warning	03/18/2025 14:13:23			DCOM	None	10016	SYSTEM	DESKTOP-BJSEBQ	Application-specific
Information	03/18/2025 14:13:25			Microsoft-Windows-Time-Service	None	37	LOCAL SERVICE	DESKTOP-BJSEBQ	The time provider NtpClient is currently receiving valid time data from time.windows.com,0x9 (ntp.r...
Information	03/18/2025 14:13:35			Microsoft-Windows-Kernel-General	11	24	LOCAL SERVICE	DESKTOP-BJSEBQ	0
Information	03/18/2025 14:13:35			Microsoft-Windows-Kernel-General	5	1	LOCAL SERVICE	DESKTOP-BJSEBQ	Possible detection of CVE-2025-03-18T08-43:35.13267602
Information	03/18/2025 14:13:35			Microsoft-Windows-Time-Service	None	35	LOCAL SERVICE	DESKTOP-BJSEBQ	The time service is now synchronizing the system time with the time source time.windows.com,0x9
Information	03/18/2025 14:13:35			Microsoft-Windows-Kernel-General	11	24	LOCAL SERVICE	DESKTOP-BJSEBQ	0
Information	03/18/2025 14:13:35			Microsoft-Windows-Kernel-General	5	1	LOCAL SERVICE	DESKTOP-BJSEBQ	Possible detection of CVE-2025-03-18T08-43:35.13308182
Information	03/18/2025 14:13:36			Microsoft-Windows-Winlogon	1101	7001	SYSTEM	DESKTOP-BJSEBQ	User Logon Notification for Customer Experience Improvement Program
Warning	03/18/2025 14:13:38			DCOM	None	10016	NETWORK SERVICE	DESKTOP-BJSEBQ	machine-default
Warning	03/18/2025 14:15:45			DCOM	None	10016	admin	DESKTOP-BJSEBQ	machine-default

To view the complete analytical information of Windows logs, click the Analytics tab

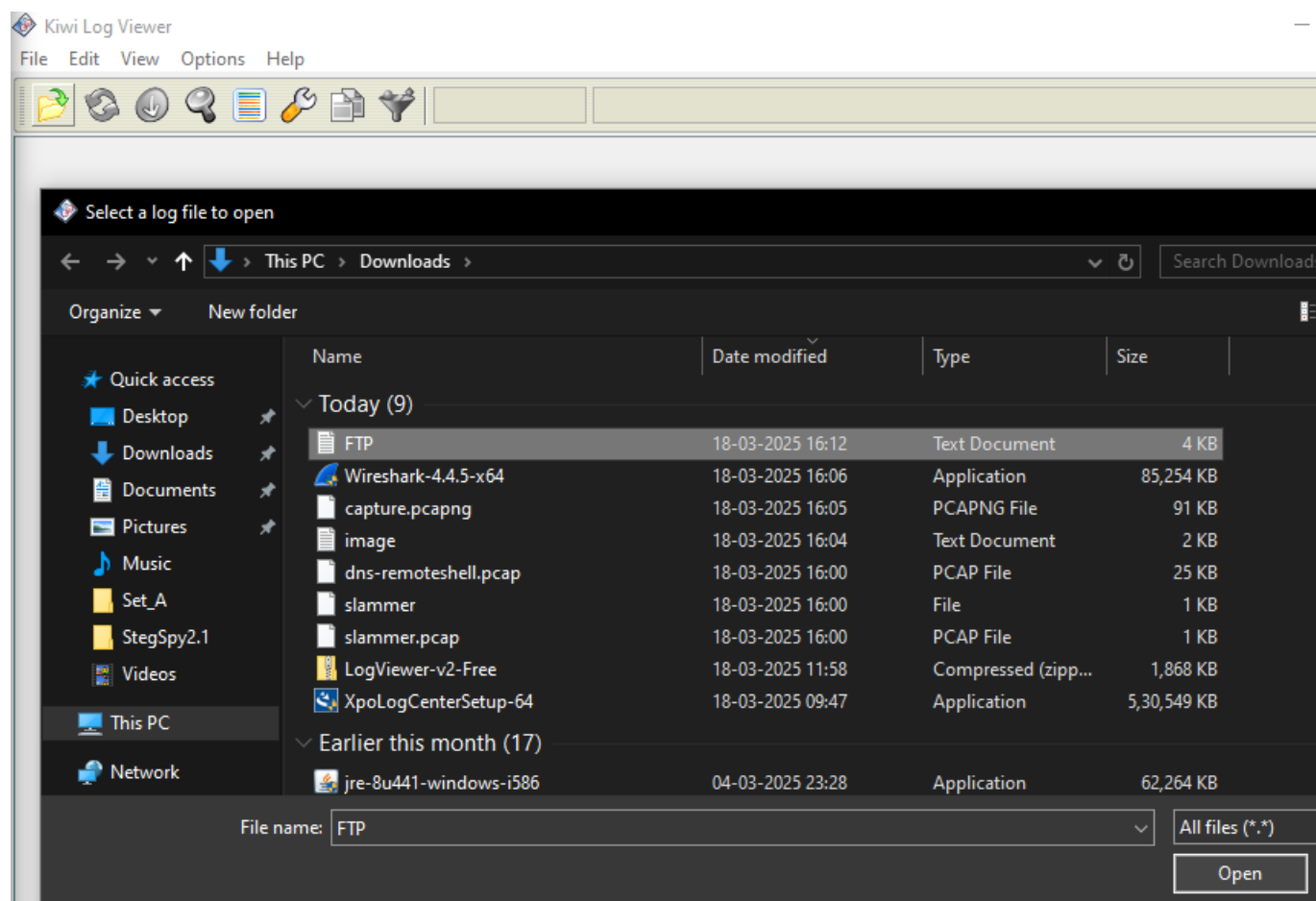


3. Investigating Network Attacks Using Kiwi Log Viewer

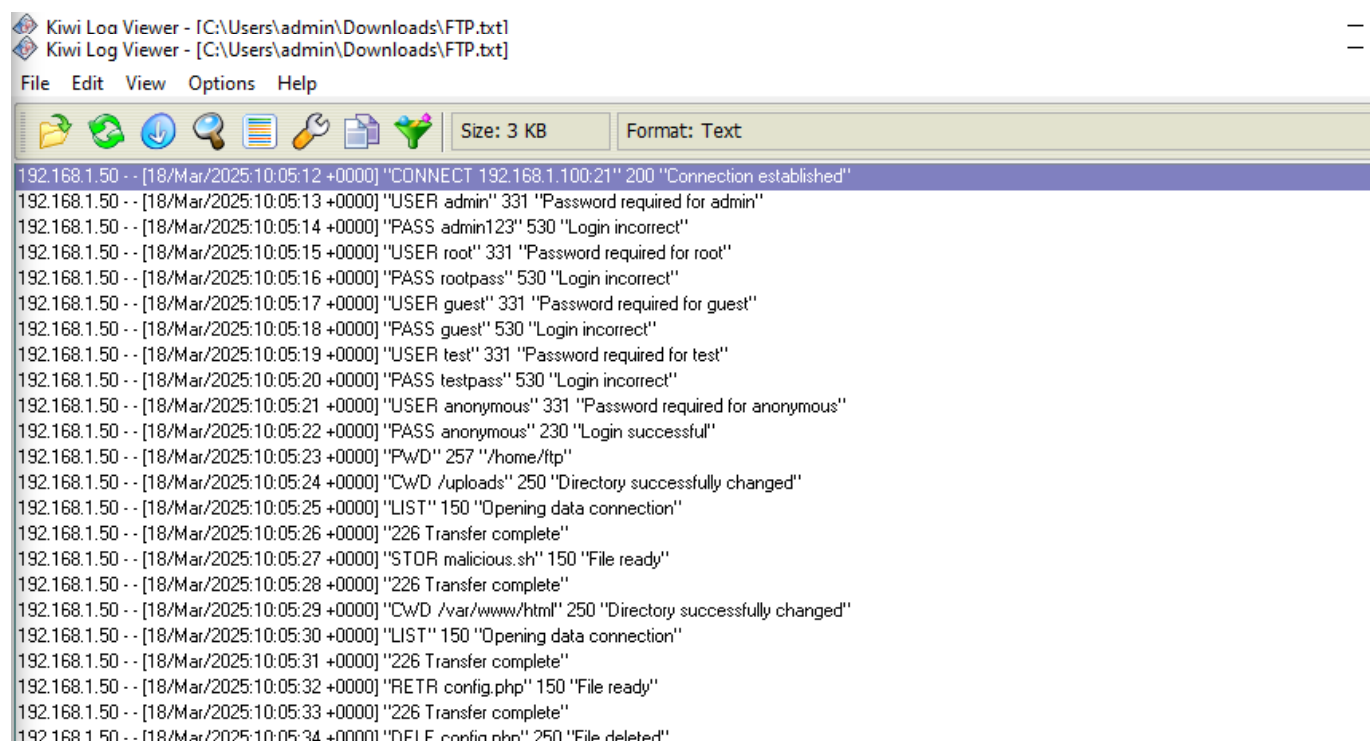
Open Kiwi Log viewer



Select a log file to open window will appear, navigate, select All files from the File Type drop-down list, select FTP and then, click Open

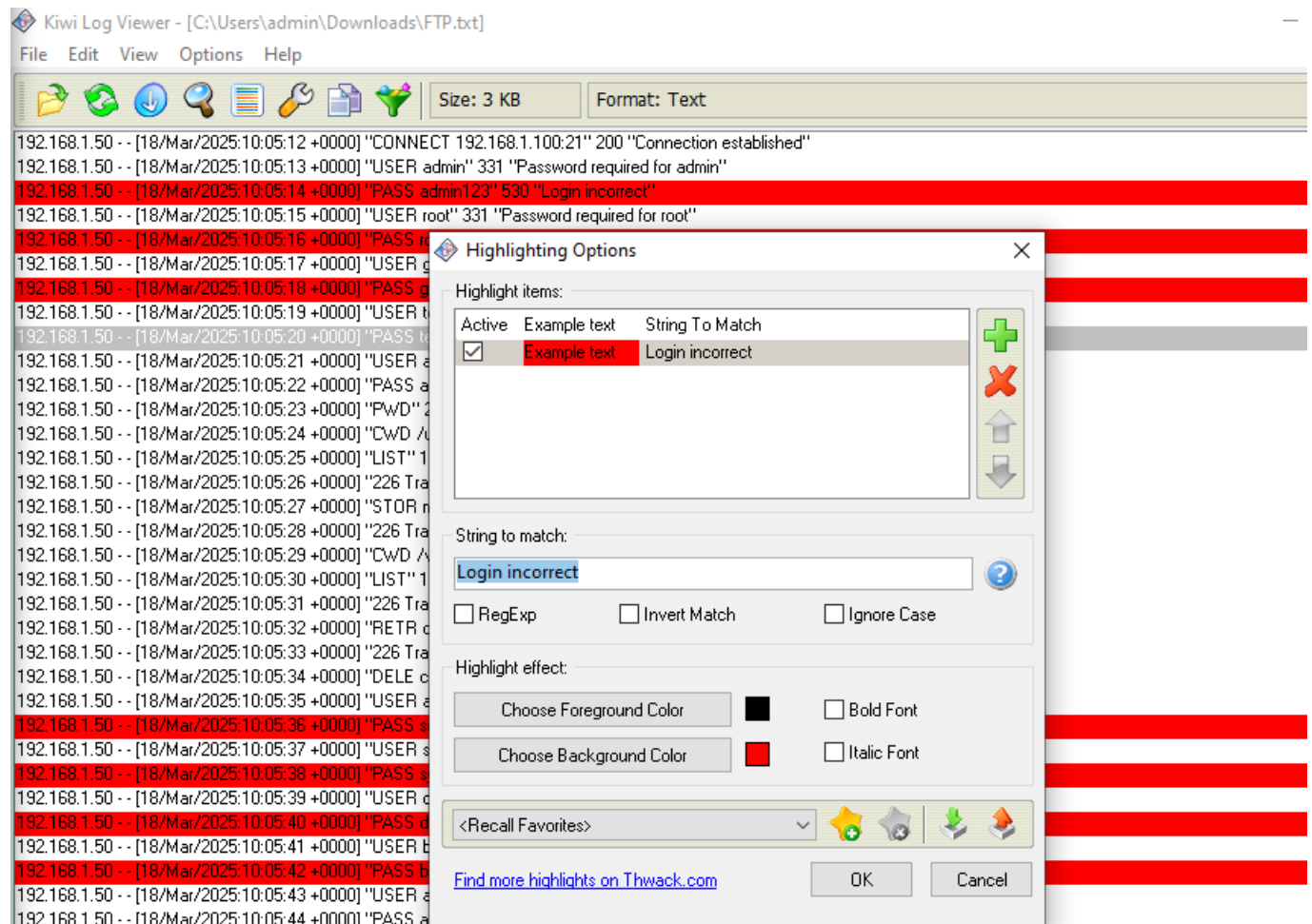


Kiwi Log Viewer displays all the logs of the selected file. You can analyze these logs to determine if there was any malicious activity in the network.

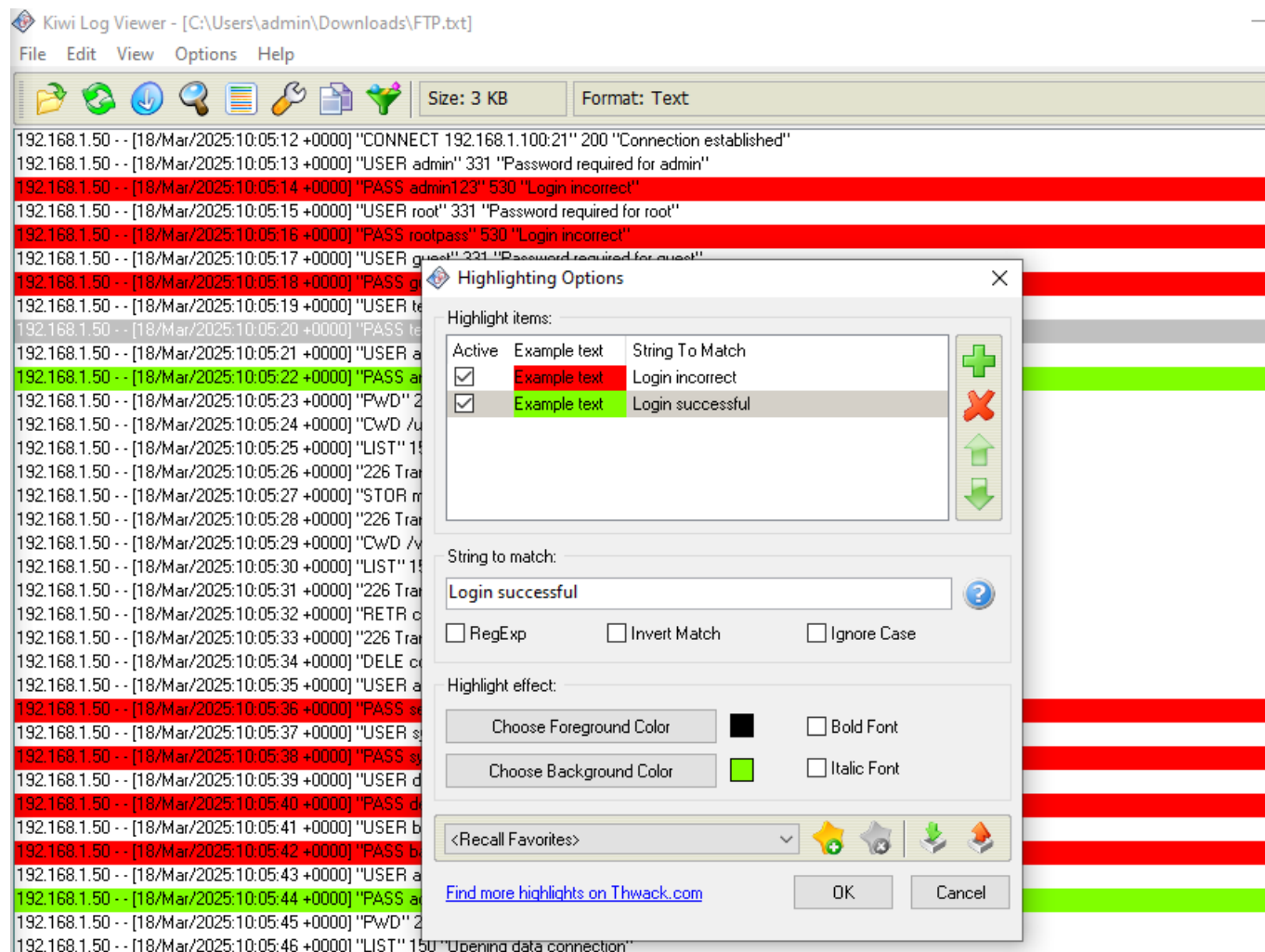


Generally, in an attempt to login to an FTP server, when a client enters invalid credentials/valid credentials/no credentials, the server returns various kinds of responses such as Response: 530, Response: 230, and Response:

To differentiate the responses, we will assign color highlights to the responses. To do so, go to Options and select Highlighting...



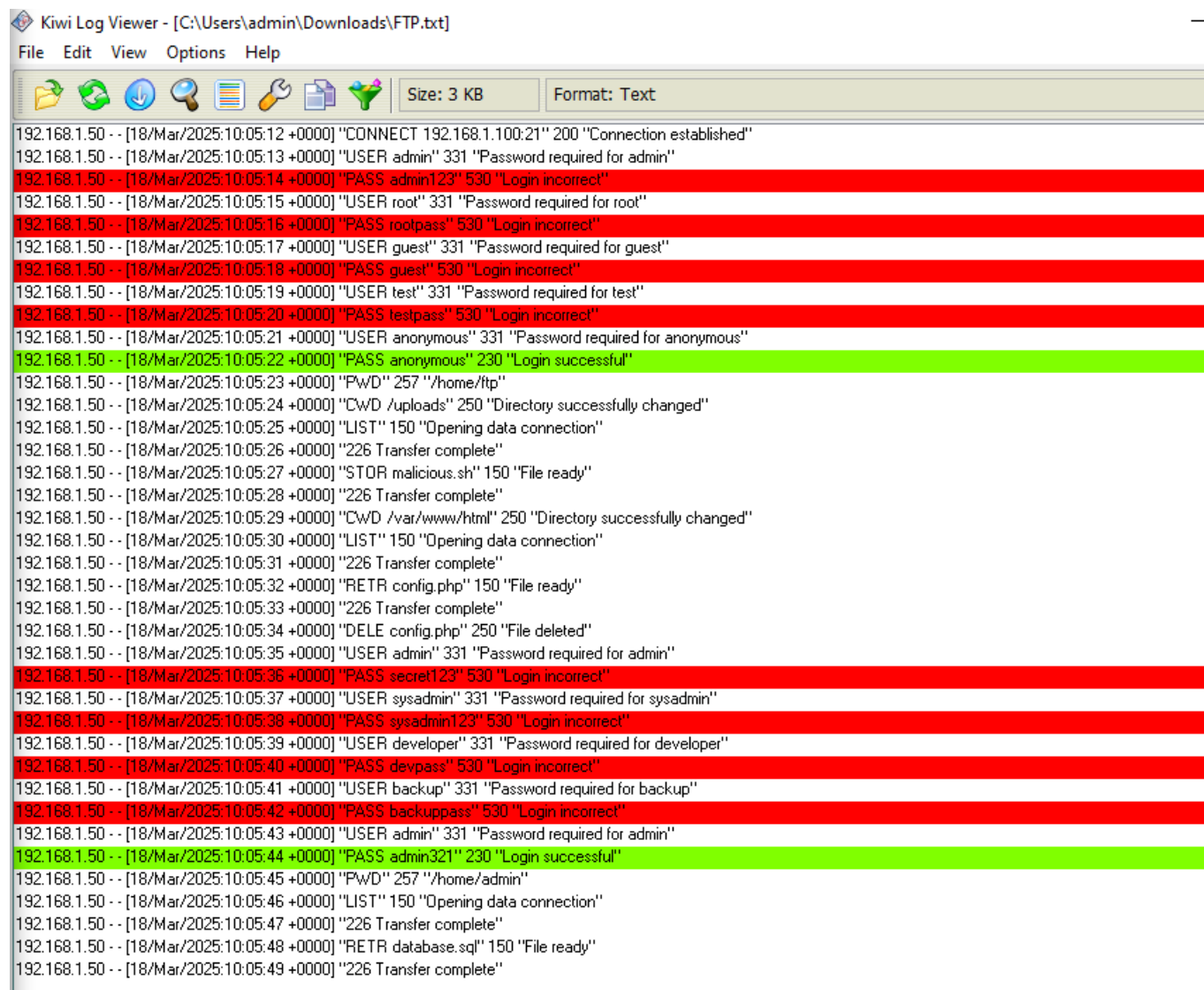
Add one more item, issue the string as Response: 230 to the item, click the Choose Background Color option from Highlight effect section, select Green color and click OK. This will highlight all the logs containing Response: 230 which represent successful login occurred by entering valid login credentials.



It is seen that there are more number of logs with red highlight, which infers that huge number of login attempts have occurred on the server, resulting in a brute force attack.

Scroll down the logs. You will observe that one of the logs (log no. 329) is highlighted in green, which means the server responded with 230 code, resulting in a successful brute-force attack."*

It provides insights into analyzing logs and identifying evidence of a brute-force attack. This green-highlighted log (no. 329) stands out as it shows a successful login response code (230) amidst numerous failed attempts (red-highlighted).

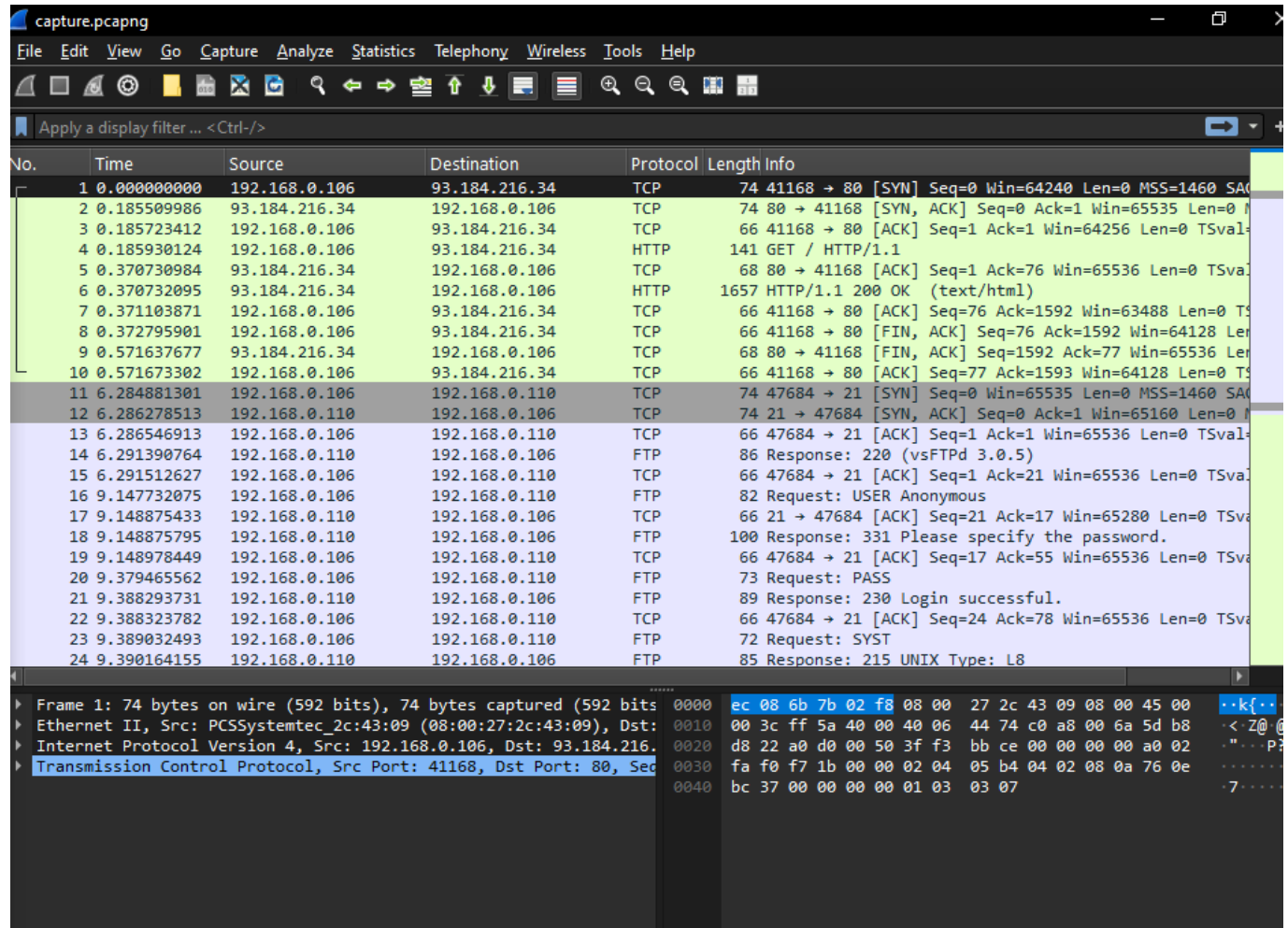


```
192.168.1.50 -- [18/Mar/2025:10:05:12 +0000] "CONNECT 192.168.1.100:21" 200 "Connection established"
192.168.1.50 -- [18/Mar/2025:10:05:13 +0000] "USER admin" 331 "Password required for admin"
192.168.1.50 -- [18/Mar/2025:10:05:14 +0000] "PASS admin123" 530 "Login incorrect"
192.168.1.50 -- [18/Mar/2025:10:05:15 +0000] "USER root" 331 "Password required for root"
192.168.1.50 -- [18/Mar/2025:10:05:16 +0000] "PASS rootpass" 530 "Login incorrect"
192.168.1.50 -- [18/Mar/2025:10:05:17 +0000] "USER guest" 331 "Password required for guest"
192.168.1.50 -- [18/Mar/2025:10:05:18 +0000] "PASS guest" 530 "Login incorrect"
192.168.1.50 -- [18/Mar/2025:10:05:19 +0000] "USER test" 331 "Password required for test"
192.168.1.50 -- [18/Mar/2025:10:05:20 +0000] "PASS testpass" 530 "Login incorrect"
192.168.1.50 -- [18/Mar/2025:10:05:21 +0000] "USER anonymous" 331 "Password required for anonymous"
192.168.1.50 -- [18/Mar/2025:10:05:22 +0000] "PASS anonymous" 230 "Login successful"
192.168.1.50 -- [18/Mar/2025:10:05:23 +0000] "PWD" 257 "/"home/ftp"
192.168.1.50 -- [18/Mar/2025:10:05:24 +0000] "CWD /uploads" 250 "Directory successfully changed"
192.168.1.50 -- [18/Mar/2025:10:05:25 +0000] "LIST" 150 "Opening data connection"
192.168.1.50 -- [18/Mar/2025:10:05:26 +0000] "226 Transfer complete"
192.168.1.50 -- [18/Mar/2025:10:05:27 +0000] "STOR malicious.sh" 150 "File ready"
192.168.1.50 -- [18/Mar/2025:10:05:28 +0000] "226 Transfer complete"
192.168.1.50 -- [18/Mar/2025:10:05:29 +0000] "CWD /var/www/html" 250 "Directory successfully changed"
192.168.1.50 -- [18/Mar/2025:10:05:30 +0000] "LIST" 150 "Opening data connection"
192.168.1.50 -- [18/Mar/2025:10:05:31 +0000] "226 Transfer complete"
192.168.1.50 -- [18/Mar/2025:10:05:32 +0000] "RETR config.php" 150 "File ready"
192.168.1.50 -- [18/Mar/2025:10:05:33 +0000] "226 Transfer complete"
192.168.1.50 -- [18/Mar/2025:10:05:34 +0000] "DELE config.php" 250 "File deleted"
192.168.1.50 -- [18/Mar/2025:10:05:35 +0000] "USER admin" 331 "Password required for admin"
192.168.1.50 -- [18/Mar/2025:10:05:36 +0000] "PASS secret123" 530 "Login incorrect"
192.168.1.50 -- [18/Mar/2025:10:05:37 +0000] "USER sysadmin" 331 "Password required for sysadmin"
192.168.1.50 -- [18/Mar/2025:10:05:38 +0000] "PASS sysadmin123" 530 "Login incorrect"
192.168.1.50 -- [18/Mar/2025:10:05:39 +0000] "USER developer" 331 "Password required for developer"
192.168.1.50 -- [18/Mar/2025:10:05:40 +0000] "PASS devpass" 530 "Login incorrect"
192.168.1.50 -- [18/Mar/2025:10:05:41 +0000] "USER backup" 331 "Password required for backup"
192.168.1.50 -- [18/Mar/2025:10:05:42 +0000] "PASS backuppass" 530 "Login incorrect"
192.168.1.50 -- [18/Mar/2025:10:05:43 +0000] "USER admin" 331 "Password required for admin"
192.168.1.50 -- [18/Mar/2025:10:05:44 +0000] "PASS admin321" 230 "Login successful"
192.168.1.50 -- [18/Mar/2025:10:05:45 +0000] "PWD" 257 "/"home/admin"
192.168.1.50 -- [18/Mar/2025:10:05:46 +0000] "LIST" 150 "Opening data connection"
192.168.1.50 -- [18/Mar/2025:10:05:47 +0000] "226 Transfer complete"
192.168.1.50 -- [18/Mar/2025:10:05:48 +0000] "RETR database.sql" 150 "File ready"
192.168.1.50 -- [18/Mar/2025:10:05:49 +0000] "226 Transfer complete"
```

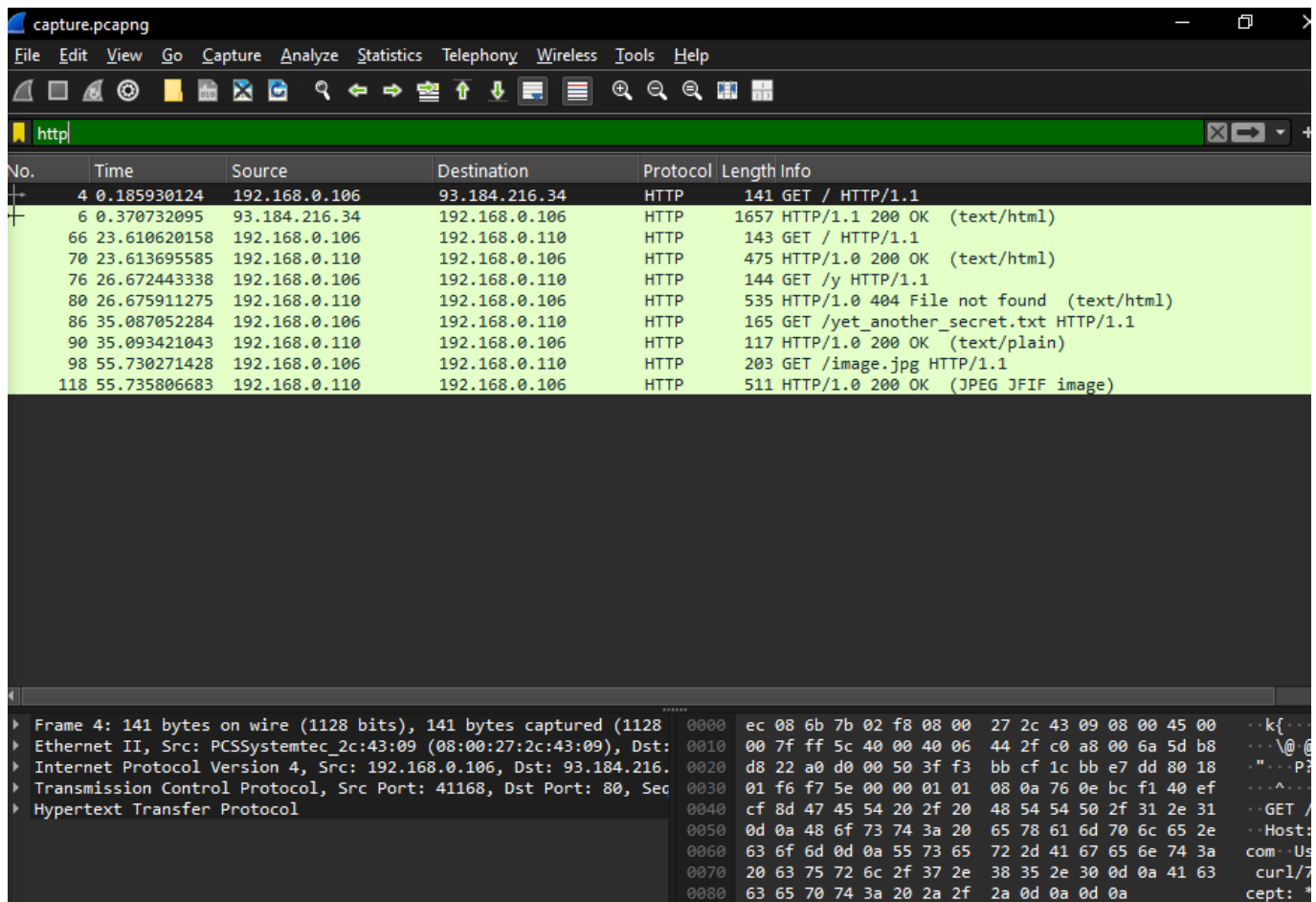
4. Investigating Network Traffic Using Wireshark

Open the pcap file in wireshark

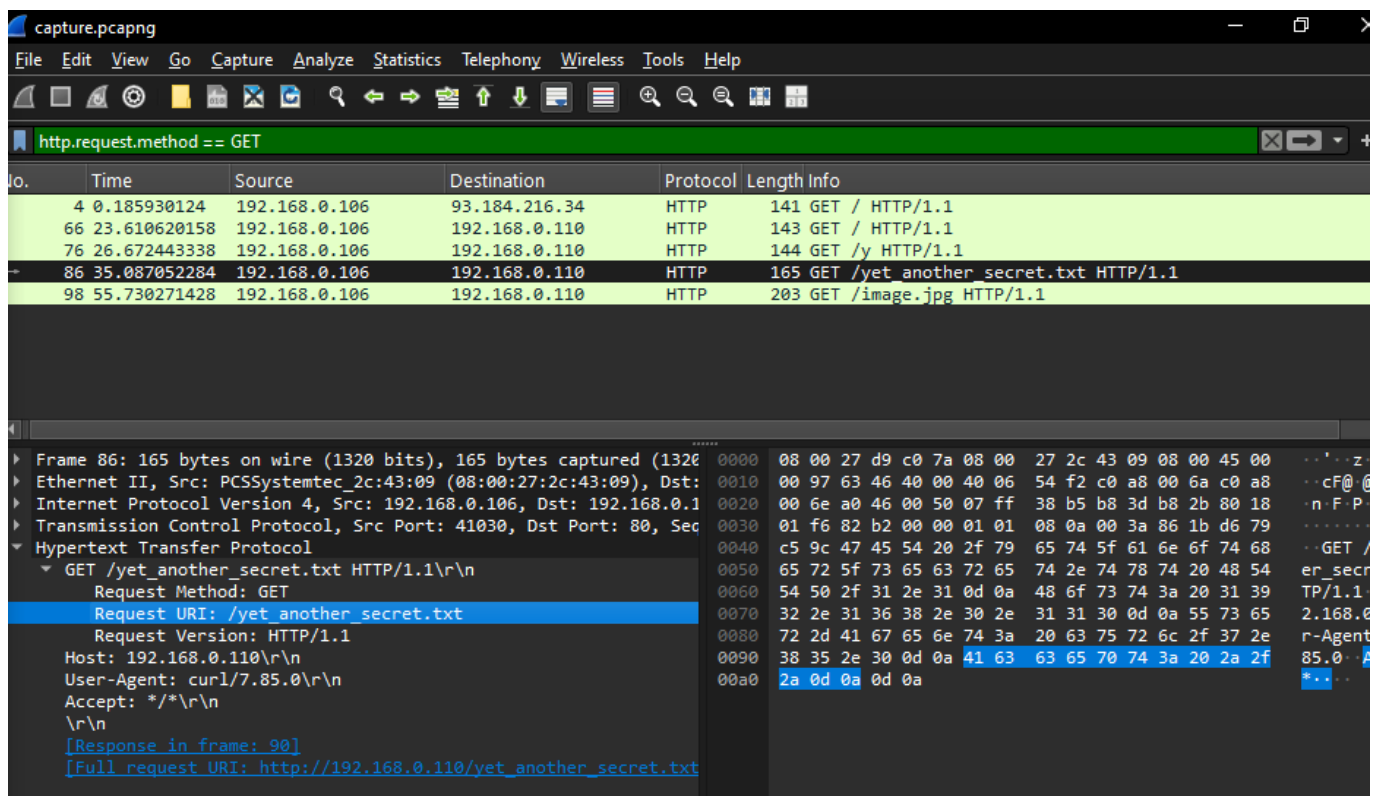
Captured packets appear in the Wireshark interface as shown in the following screenshot.



Type http in the Filter field and press Enter to filter the HTTP traffic.

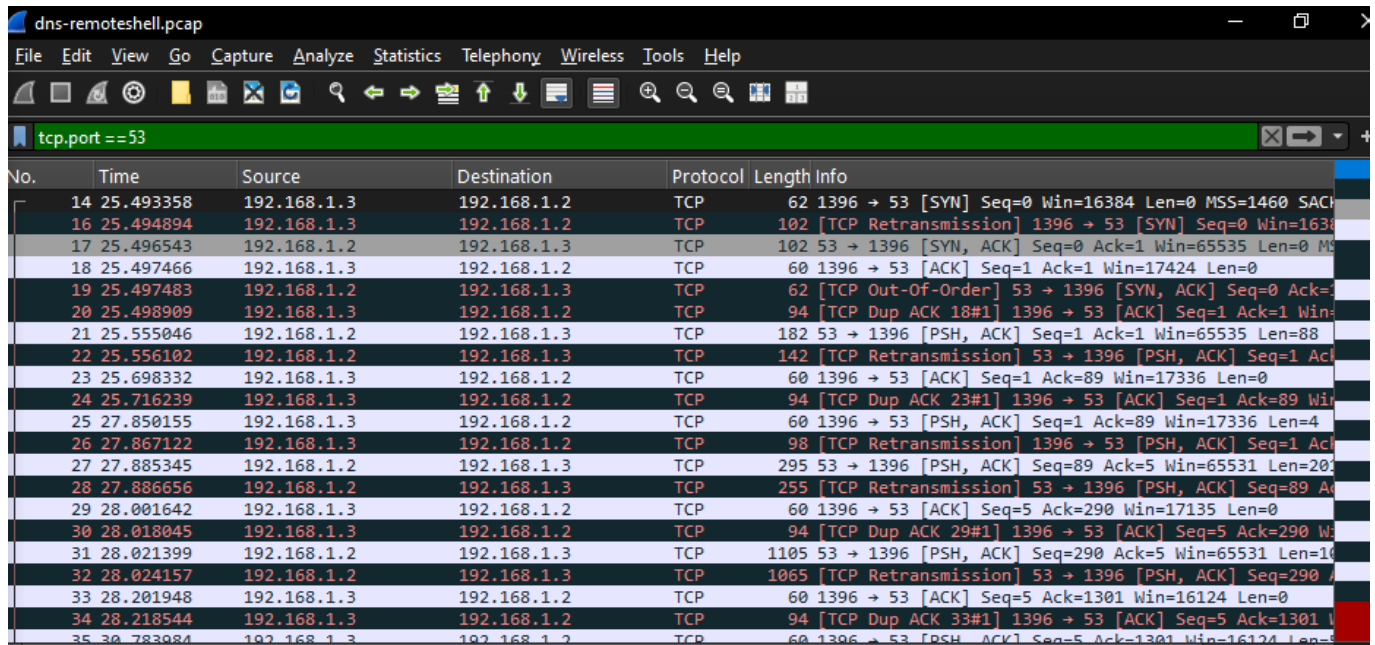


Type the filter `http.request.method == GET` in Filter field and press Enter. Wireshark filters the traffic containing GET requests and displays them as shown in the following screenshot



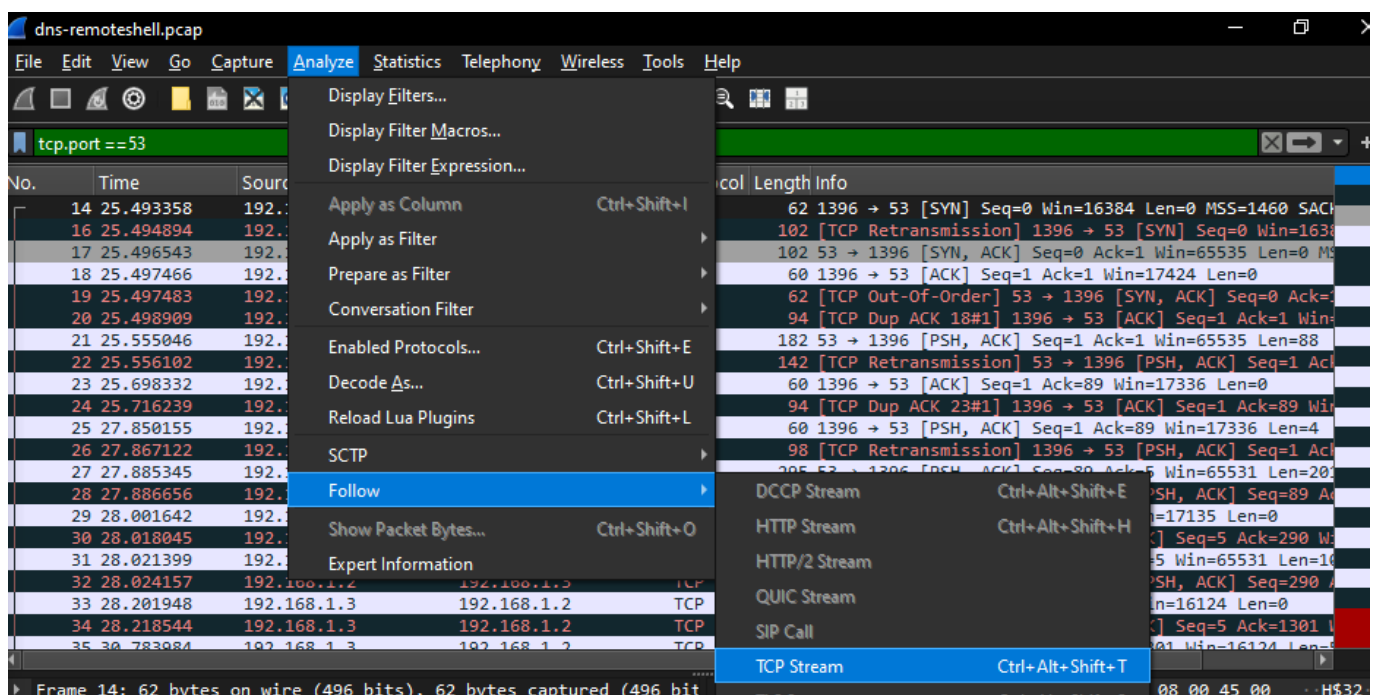
Now, we shall look for DNS Anomalies in the network. Close the current packet capture file in Wireshark

Since DNS uses port 53 for communication, we shall be filtering the traffic flowing on port number 53. To filter, type the command `tcp.port == 53` in the Filter field and press Enter. Wireshark filters the traffic flowing on port 53 and displays it as shown in the following screenshot



To view the data in a sequence, we will use Follow TCP Stream option in Wireshark. Now, you need to examine the data flowing through these packets. To view the data in a sequence, we will use Follow TCP Stream option in Wireshark.

Right-click on any one of the packets between 14 and 38 (here, packet 16), select Follow and click TCP Stream from the drop-down list.



You can observe that a remote shell has been established on port 53, and the directory listing has been performed on the remote machine; which is evident from the following screenshot.

```
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\>
dir
dir
Volume in drive C has no label.
Volume Serial Number is FF47-80EB

Directory of C:\

01/12/2005  11:59 AM                0 aierrorlog.txt
01/19/2004  09:45 PM                0 AUTOEXEC.BAT
01/19/2004  09:45 PM                0 CONFIG.SYS
06/26/2004  12:12 PM        <DIR>      Documents and Settings
02/03/2005  11:40 PM        <DIR>      EasyBoot
02/29/2004  02:51 PM       11,531 installer-debug.txt
12/19/2004  12:50 AM        <DIR>      mga
12/19/2004  12:51 AM        <DIR>      mgafold
11/24/2004  07:47 PM        <DIR>      mnt
10/07/2004  10:01 AM        <DIR>      movie
06/26/2004  01:03 PM        <DIR>      My Downloads
01/13/2005  10:52 PM        <DIR>      Program Files
01/04/2005  10:27 AM        <DIR>      quarantine
04/19/2004  09:57 PM       7,241 s37g
10/31/2004  08:36 PM                0 s3fs
06/02/2004  08:54 PM       123 systemscandata.txt
08/08/2004  10:48 AM        <DIR>      Temp
12/12/2004  02:24 PM     94,135,944 temp.mpg
01/13/2005  06:10 PM        <DIR>      WINDOWS
11/20/2004  09:27 AM        <DIR>      WUTemp
           8 File(s)      94,154,839 bytes
          12 Dir(s)      7,145,897,984 bytes free

C:\>
exit
```